Subject: GEOTECHNICAL ENGINEERING LAB

Subject Code: 23A01503P

LABORATORY MANUAL

Geotechnical Engineering Lab (B.Tech Civil, JNTUA R23)

Course Objectives

- Understand the fundamental index properties of soils and their significance in geotechnical engineering.
- Perform field and laboratory tests to determine in-situ density and compaction characteristics of soils.
- Evaluate permeability, shear strength and consolidation of soil.
- Analyse strength and deformation of soils through shear and compression tests.
- Interpret test results and relate them to real-world geotechnical design problems.

Course Outcomes

- **Determine index properties** specific gravity, grain size distribution and consistency limits.
- Conduct field and laboratory compaction tests to get the moisture-density relationship.
- Evaluate permeability and consolidation using suitable laboratory techniques.
- Analyse shear strength and compressibility by direct shear, unconfined compression and tri-axial tests.
- Integrate test results with engineering judgement for geotechnical applications.

Note any 10 of the following experiments are to be done as per the syllabus. All IS are given here.

List of Experiments

No	Experiment
1a	Specific Gravity of Soil
1b	Grain Size Distribution - Sieve Analysis
1c	Grain Size Distribution - Hydrometer Analysis
1d	Liquid Limit and Plastic Limit Tests
1e	Shrinkage Limit and Differential Free Swell Tests
2a	Field Density Test (Sand Replacement Method)
2b	Moisture-Density Relationship by Standard Proctor Compaction Test
3a	Permeability - Constant Head Method
3b	Permeability - Falling Head Method
3c	Coefficient of Consolidation
3d	Direct Shear Test in Cohesionless Soil
3e	Unconfined Compression Test in Cohesive Soil
3f	Laboratory Vane Shear Test in Cohesive Soil
3g	Tri-Axial Compression Test in Cohesionless Soil
3h	California Bearing Ratio (CBR) Test

Experiment 1a - Specific Gravity of Soil

Aim

To determine the specific gravity of soil solids using a pycnometer (density bottle). Reference: IS 2720 (Part 3).

Apparatus

- Pycnometer (500 ml) with brass cap, balance (0.01 g accuracy), oven (105 to 110 C), distilled water, glass rod, sieve 4.75 mm, vacuum pump or hot plate.

Theory

Specific gravity G is the ratio of the mass of a given volume of soil solids to the mass of an equal volume of water at the same temperature. Typical values: sand 2.65 to 2.67, silt 2.67 to 2.70, clay 2.70 to 2.80, organic soil below 2.0.

Formula $G = (W2 - W1) / [(W2 - W1) - (W3 - W4)]$

Procedure

- Clean and dry the pycnometer; weigh it with cap = W1.
- Put about 200 g of oven dried soil (passing 4.75 mm) in it and weigh = W2.
- Add distilled water to cover the soil, stir with the glass rod and remove entrapped air by shaking or vacuum for about 10 minutes.
- Fill the pycnometer with water to the top mark, dry outside and weigh = W3.
- Empty, wash, fill with distilled water only up to the mark and weigh = W4.
- Repeat for 2 or 3 trials and take the average.

Observations

Trial	W1 (g)	W2 (g)	W3 (g)	W4 (g)	G
1	152.97	202.69	682.81	651.92	2.64
2	152.22	202.25	681.61	650.32	2.67
3	152.17	202.18	680.42	649.28	2.65

Result and Precautions

- Specific gravity of the soil G = 2.65 (average).
- Remove all air bubbles; keep temperature constant; use distilled water; dry the outside before weighing.

Experiment 1b – Grain Size Distribution by Sieve Analysis

Aim

To determine the particle size distribution of coarse grained soil (retained on 75 micron) and to find D10, D30, D60, Cu and Cc. Reference: IS 2720 (Part 4).

Apparatus

- Set of IS sieves (4.75 mm, 2 mm, 1 mm, 600, 425, 300, 212, 150, 75 micron) with lid and pan, sieve shaker, balance, oven, brush.

Theory

Sieve analysis separates soil particles by size. The percent finer is plotted against the log of particle size to get the grain size distribution curve. From it the soil is classified as well graded or poorly graded.

Formulas Percent retained = (weight retained / total weight) x 100. Cumulative percent retained = sum of percent retained. Percent finer N = 100 - cumulative percent retained. Uniformity coefficient Cu = D60 / D10. Coefficient of curvature Cc = (D30 squared) / (D60 x D10).

Procedure

- Take 500 g to 1000 g of oven dried soil; break lumps without crushing grains.
- Arrange the sieves in order with the largest on top and the pan at bottom.
- Pour the soil on the top sieve, cover it and shake for 10 minutes using the shaker.
- Weigh the soil retained on each sieve and the pan.
- Compute percent retained, cumulative percent and percent finer; plot the semi-log curve.

Observations

IS sieve size	Wt retained (g)	% retained	Cumulative % retained	% finer
4.75 mm	12	2.4	2.4	97.6
2.0 mm	38	7.6	10.0	90.0

1.0 mm	74	14.8	24.8	75.2
600 micron	96	19.2	44.0	56.0
425 micron	84	16.8	60.8	39.2
300 micron	70	14.0	74.8	25.2
212 micron	60	12.0	86.8	13.2
150 micron	38	7.6	94.4	5.6
75 micron	20	4.0	98.4	1.6
Pan	8	1.6	100.0	0.0
Total	500	100.0		

Result

- D10 = 0.18 mm, D30 = 0.34 mm, D60 = 0.67 mm, Cu = 3.6, Cc = 0.93.
- Well graded if Cu is greater than 4 (gravel) or 6 (sand) and Cc is between 1 and 3; otherwise poorly graded.
- **Precaution** total loss of weight should not exceed 1 percent; do not overload the sieves.
- Here Cu = 3.6 (less than 6) so the sand is poorly graded (uniformly graded).

Experiment 1c – Grain Size Distribution by Hydrometer Analysis

Aim

To find the particle size distribution of fine grained soil (passing 75 micron) by hydrometer method. Reference: IS 2720 (Part 4).

Apparatus

- Hydrometer (152 H), two measuring jars of 1000 ml, dispersing agent (sodium hexametaphosphate), mechanical stirrer, thermometer, stop watch, balance.

Theory

Based on Stokes law, which gives the velocity of a sphere settling in a liquid. The diameter of the particle that has settled to the hydrometer depth after time t is

Formula D = sqrt[30 x viscosity x H / ((G - Gw) x t)] (D in mm, t in minutes). Or D = K x sqrt(Hr / t), where Hr is the effective depth and K depends on G and temperature.

- Percent finer N = (R x G) / (W x (G - 1)) x 100 (after corrections), where R is corrected hydrometer reading and W is dry soil weight.

Procedure

- Take 50 g of oven dried soil passing 75 micron; mix with 100 ml of dispersing agent solution and soak overnight.
- Stir the mixture in the mechanical stirrer for 15 minutes and transfer to the jar; make up to 1000 ml with distilled water.
- Shake the jar well, place it on a table and start the stop watch.
- Insert the hydrometer and note readings at 0.5, 1, 2, 4, 8, 15, 30 minutes, 1, 2, 4, 8 and 24 hours; remove the hydrometer after each reading.
- Note the temperature. Apply the meniscus, temperature and dispersing agent corrections.
- Compute D and percent finer and plot the curve.

Observations

Time t (min)	Hydrometer reading Rh	Corrected R	Effective depth Hr (cm)	D (mm)	% finer N
0.5	31	30	11.4	0.0596	96.4
1	29	28	11.7	0.0428	89.9
2	26	25	12.2	0.0309	80.3
4	23	22	12.7	0.0223	70.7

Result and Precautions

- Percent of silt and clay sizes read from the curve = 70.7 % finer than 0.0223 mm (at 4 min).
- Take care not to disturb the suspension; keep the temperature constant;

correct readings properly.

Experiment 1d – Liquid Limit and Plastic Limit Tests

Aim

To determine the liquid limit (wL), plastic limit (wP), plasticity index (Ip) and to classify the soil. Reference: IS 2720 (Part 5).

Apparatus

- Casagrande liquid limit device, grooving tool, spatula, evaporating dish, glass plate, 3 mm rod, moisture cans, oven, balance, wash bottle, sieve 425 micron.

Theory

Consistency limits (Atterberg limits) mark the change of the soil state from liquid to plastic to semi-solid. Liquid limit is the water content at which a standard groove closes by 12 mm after 25 blows. Plastic limit is the water content at which the soil just begins to crumble when rolled into 3 mm thread.

Formulas Plasticity index Ip = wL - wP. Flow index If = (w1 - w2) / log10(N2 / N1). Toughness index It = Ip / If. Consistency index Ic = (wL - w) / Ip.

Procedure – Liquid limit

- Take 120 g of soil passing 425 micron; mix with water to a uniform paste.
- Place a portion in the cup, level it, cut a groove with the tool and turn the handle at 2 revolutions per second.
- Count the blows until the groove closes by 12 mm. Take a sample from the closed part for water content.
- Repeat with different water contents to get 4 readings between 10 and 40 blows.
- Plot water content (natural) against log N (flow curve); the water content at 25 blows is liquid limit.

Procedure – Plastic limit

- Take 30 g of soil and mix with water until it is plastic enough to roll.
- Roll it on the glass plate with the palm to make a thread. When the thread of 3 mm diameter starts to crack, collect it and find its water content. Repeat 3 times; the average is plastic limit.

Observations

Trial	No. of blows N	Can wt (g)	Wet soil + can (g)	Dry soil + can (g)	Water content %
1	14	14.59	42.91	32.47	58.4
2	21	14.28	45.00	34.11	54.9
3	27	14.94	42.54	33.12	51.8
4	34	14.40	45.66	35.35	49.2

Plastic limit observations:

PL trial	Can wt (g)	Wet soil + can (g)	Dry soil + can (g)	Water content %
1	15.05	27.14	24.78	24.3
2	14.56	28.23	25.49	25.1
3	14.24	27.43	24.82	24.7

Result

- wL = 52.6 %, wP = 24.7 %, Ip = 28.0 %. If Ip is more than 17 the soil is highly plastic; between 7 and 17 medium plastic; less than 7 low plastic.
- **Precaution** do not add dry soil to a wet paste; avoid sliding of the paste in the cup; keep the base of the apparatus clean.
- Here Ip = 28.0 %, which is more than 17, so the soil is highly plastic (clay of high plasticity, CH).

Experiment 1e – Shrinkage Limit and Differential Free Swell

Aim

(a) To determine the shrinkage limit and shrinkage ratio. (b) To find the differential free swell index (DFS). Reference: IS 2720 (Parts 6 and 40).

Apparatus

- Shrinkage dish, mercury or wax, glass cup, glass plate with prongs, balance, oven, graduated cylinders (100 ml, two), kerosene, distilled water, sieve 425 micron.

Theory

- Shrinkage limit is the water content below which no further volume change happens when the soil is dried.
- Free swell indicates the expansive nature of the soil (black cotton soil).

Formulas Shrinkage limit ws = [(M1 - M2)/M2 x 100] - [(V1 - V2) x rw / M2 x 100]. Shrinkage ratio SR = M2 / (V2 x rw). DFS = [(Vd - Vk) / Vk] x 100. Here M1 is wet soil mass, M2 dry mass, V1 initial volume, V2 dry volume, Vd is volume in distilled water and Vk is volume in kerosene.

Procedure - Shrinkage limit

- Prepare a wet paste at about the liquid limit; fill the shrinkage dish (coated with grease) in 3 layers, tapping to remove air; level and weigh = M1; note dish volume V1.
- Dry in air and then in oven; weigh the dry pat = M2.
- Find the volume of the dry pat V2 by displacement of mercury in the glass cup.

Procedure - Differential free swell

- Take two samples of 10 g oven dried soil passing 425 micron.
- Put one in a 100 ml cylinder with distilled water and the other with kerosene; stir to remove air, make up to 100 ml.
- Leave for 24 hours and read the settled volumes Vd and Vk.

Observations and Result

Item	Value
M1, M2 (g)	30.2, 20.5
V1, V2 (cm3)	16.4, 11.8
Shrinkage limit ws (%)	24.9
Vd, Vk (ml)	17, 13
DFS (%)	30.8

- DFS below 20 percent is low swelling, 20 to 35 moderate, 35 to 50 high, above 50 very high degree of expansion.
- Shrinkage ratio SR = M2 / V2 = 1.74. DFS = 30.8 % so the soil has moderate swelling.

Experiment 2a - Field Density by Sand Replacement Method

Aim

To determine the in-situ dry density of soil using sand replacement method.

Reference: IS 2720 (Part 28).

Apparatus

- Sand pouring cylinder (3 litre) with conical funnel and shutter, calibrating can, metal tray with central hole (300 mm), tools for digging, balance, standard (Ottawa or Calcutta) sand passing 600 micron and retained on 300 micron, moisture cans, oven.

Theory

The volume of the hole excavated in the soil is found by filling it with sand of known density; the mass of excavated soil gives the bulk density.

Formulas Bulk density gamma = Ws / V. Dry density gamma_d = gamma / (1 + w). Volume of hole V = W_sand in hole / gamma_sand. Sand in hole = W1 - W2 - Wc, where W1 is initial mass of cylinder with sand, W2 final mass and Wc is mass of sand in the cone (found by calibration).

Procedure

- **Calibrate** find the bulk density of sand by filling the calibrating can of known volume with sand from the cylinder; find the mass of sand filling the cone by running sand on a plain surface.
- Clean and level the test place, fix the tray, dig a hole of about 100 mm depth through the tray hole, collect all the soil and weigh = Ws.
- Place the sand pouring cylinder over the hole, open the shutter, allow the sand to fill the hole and cone and close the shutter. Weigh the cylinder = W2.
- Find the water content of the dug soil.

Observations

Item	Value
Mass of cylinder + sand before W1 (g)	6850
Mass after W2 (g)	3580
Mass of sand in cone Wc (g)	1330
Mass of sand in hole (g)	1940
Density of sand (g/cm3)	1.52
Volume of hole (cm3)	1276
Mass of wet soil Ws (g)	2480
Water content w (%)	12.6

Result

- Bulk density = 1.94 g/cm3, dry density = 1.73 g/cm3. Compare with the laboratory maximum dry density to find the degree of compaction (usually 95 to 98 percent is required). With MDD = 1.79 g/cm3 from the Proctor test,

degree of compaction = 96.4 %.

- **Precaution** do not vibrate the cylinder when sand is running; avoid loss of soil during digging; the hole should have a smooth side.

Experiment 2b - Standard Proctor Compaction Test

Aim

To determine the optimum moisture content (OMC) and the maximum dry density (MDD) of soil. Reference: IS 2720 (Part 7).

Apparatus

- Proctor mould 1000 cc (100 mm dia, 127.3 mm height) with base plate and collar, rammer 2.6 kg with 310 mm free fall, balance, sieve 20 mm and 4.75 mm, straight edge, mixing tray, moisture cans, oven, measuring jar.

Theory

Compaction is the densification of soil by mechanical energy. For a given effort the dry density is maximum at one particular water content (OMC). Energy per unit volume = 592 kJ/m3 for the standard test (3 layers, 25 blows each).

Formulas Bulk density gamma = (W2 - W1) / V. Dry density gamma_d = gamma / (1 + w). Zero air void line :: gamma_d = G gamma_w / (1 + wG).

Procedure

- Take 3 kg of air dried soil passing 4.75 mm (or 20 mm); add water, say 8 percent for sand and 12 percent for clay, and mix well.
- Weigh the empty mould with base = W1. Fix the collar; fill soil in 3 layers, giving 25 blows of the rammer on each layer, evenly distributed.
- Remove the collar, trim the top level and weigh = W2.
- Take a sample from the middle for water content.
- Repeat with increasing water content by 2 percent each time for at least 5 trials until the weight starts to fall.

- Plot dry density against water content; the peak gives MDD and OMC.

Observations

Trial	Water added %	W1 (g)	W2 (g)	Bulk density	Water content w	Dry density
1	8	4120	22400	18.28	8.3	16.88
2	10	4120	22500	18.38	10.2	16.68
3	12	4120	22590	18.47	12.4	16.43
4	14	4120	22640	18.52	14.1	16.23
5	16	4120	22635	18.52	16.3	15.92

Result

- OMC = 14.0 %, MDD = 1.79 g/cm3. Plot the zero air void line to check the curve lies to its left.
- **Precaution** use equal blows and uniform fall; do not reuse soil in the next trial; break lumps without crushing grains.

Experiment 3a - Permeability by Constant Head Method

Aim

To determine the coefficient of permeability of coarse grained soil. Reference: IS 2720 (Part 17).

Apparatus

- Permeameter mould (100 mm dia, 127.3 mm height) with drainage base and cap, constant head tank, measuring jar, stop watch, thermometer, scale.

Theory

Darcy's law $v = k \times i$, so $Q = k \times i \times A$. In the constant head test the head difference h is kept constant across the soil.

Formula $k = Q\,L / (A\,h\,t)$ where Q = volume of water collected in time t, L = length of sample, A = area, h = constant head. k at 27 C = k_T x (viscosity at T / viscosity at 27 C).

Procedure

- Prepare the soil sample at required density in the mould; place porous stones and filter papers and fix the cap.
- Saturate the sample by letting water flow up through it.
- Connect to the constant head supply and allow flow to become steady.
- Collect the water for a fixed time (say 60 seconds) in the jar and measure it. Repeat 3 times with different heads. Note the temperature.

Observations

Trial	Head h (cm)	Time t (s)	Discharge Q (cm3)	k (cm/s)
1	30	60	355	0.0320
2	40	60	459	0.0310
3	50	60	611	0.0330

Result

- **k = 0.0320 cm/s (at 27 C). Typical k** gravel above 1, sand 1 to 0.001, silt 0.001 to 0.00001, clay less than 0.000001 cm/s.
- **Precaution** remove air bubbles; maintain constant head; avoid leakage along the sides.

Experiment 3b - Permeability by Falling Head Method

Aim

To determine the coefficient of permeability of fine grained soils (silt and clay). Reference: IS 2720 (Part 17).

Apparatus

- Permeameter with standpipe of known area a, stop watch, thermometer, water supply, scale.

Theory

Here the water level in the standpipe falls from h1 to h2 in time t.

Formula k = [2.303 a L / (A t)] x log10 (h1 / h2) where a = area of standpipe, A = area of sample, L = length of sample.

Procedure

- Prepare and saturate the sample as in the constant head test.
- Fill the standpipe with water up to a convenient level h1, start the stop watch and note the time t for the level to fall to h2.
- Repeat several times with different h1 and h2 and take the mean k. Note the temperature and correct to 27 C.

Observations

Trial	h1 (cm)	h2 (cm)	Time t (s)	k (cm/s)
1	100	80	6310	4.50e-06
2	80	62	6902	4.70e-06
3	100	78	7186	4.40e-06

Result

- **k = 4.53e-06 cm/s. Precaution** the sample should be fully saturated; use deaired water; check that the connections are leak free.
- Data used: standpipe area a = 0.785 cm2, sample area A = 78.54 cm2, sample length L = 12.73 cm.

Experiment 3c - Coefficient of Consolidation

Aim

To determine the coefficient of consolidation cv and the compression index of a soil by the one dimensional consolidation test. Reference: IS 2720 (Part 15).

Apparatus

- Consolidometer (oedometer) with fixed or floating ring (60 mm dia, 20 mm height), porous stones, loading frame with weights, dial gauge (0.002 mm), stop watch, water container.

Theory

When a saturated soil is loaded, water comes out slowly and the volume decreases with time. Terzaghi's theory gives the time factor Tv relating time and the degree of consolidation.

Formulas Coefficient of consolidation cv = Tv x Hdr squared / t. For 90 percent consolidation Tv = 0.848 (root time method); for 50 percent Tv = 0.197 (log time method). Compression index Cc = (e0 - e1) / log10 (p1 / p0). Coefficient of volume change mv = (delta e / (1 + e0)) / delta p. Permeability k = cv x mv x gamma_w.

Procedure

- Cut the sample into the ring; weigh it and find its initial water content and void ratio.
- Place the ring with porous stones in the cell, fill water and set the dial gauge.
- Apply the first load (seating load of 0.05 kg/cm2). After 24 hours apply loads in steps of 0.1, 0.2, 0.5, 1, 2, 4 kg/cm2, doubling each time.
- For each load take dial readings at 0.25, 1, 2.25, 4, 6.25, 9, 16, 25, 36, 49, 64, 100, 200 minutes and 24 hours.
- Plot dial reading against square root of time (Taylor) or log time (Casagrande) and find t90 or t50.
- Plot the e - log p curve to get Cc.

Observations

Load (kg/cm2)	Final dial reading (mm)	Change in thickness (mm)	Void ratio e	t90 (min)	cv (cm2/s)
0.1	0.18	0.18	0.932	9.0	0.00156
0.2	0.36	0.18	0.915	7.6	0.00181
0.5	0.78	0.42	0.874	6.1	0.00219
1.0	1.34	0.56	0.819	4.8	0.00264
2.0	2.02	0.68	0.753	4.1	0.00289

Result

- **cv = 0.00222 cm2/s, Cc = 0.201. Precaution** avoid sample disturbance; keep the water level above the sample; do not jerk the apparatus.
- Data used: initial sample thickness 20 mm, diameter 60 mm, initial void ratio e0 = 0.95, double drainage.

Experiment 3d - Direct Shear Test (Cohesionless Soil)

Aim

To determine the shear strength parameters (angle of internal friction phi) of sand by direct shear test. Reference: IS 2720 (Part 13).

Apparatus

- Direct shear box (60 mm x 60 mm) with grid plates, porous stones, loading yoke and hanger, proving ring, horizontal and vertical dial gauges, motorised loading unit, tamper, balance.

Theory

Coulomb's equation shear strength s = c + sigma x tan phi. For sand c = 0, so s = sigma tan phi. Tests are done under 3 different normal stresses and the peak shear stress is plotted against normal stress.

Formulas Normal stress sigma = N / A. Shear stress tau = proving ring reading x constant / A (corrected area). phi = tan inverse (tau / sigma).

Procedure

- Assemble the box, fix the grid plate with serrations perpendicular to the direction of shear and place the sand at the required density.
- Put the upper grid, loading pad and the loading yoke; apply the normal load, e.g. 0.5, 1.0 and 1.5 kg/cm2.
- Remove the pins joining the two halves, set the dial gauges to zero and shear at a rate of 1.25 mm per minute.
- Record the proving ring and horizontal dial readings every 0.2 to 0.25 mm until failure or 20 percent strain.
- Repeat for other normal stresses with fresh samples.

Observations

Test	Normal stress (kg/cm2)	Maximum shear force (kg)	Shear stress at failure (kg/cm2)
1	0.5	11.6	0.322
2	1.0	23.6	0.656
3	1.5	35.2	0.978

Result

- Plot shear stress vs normal stress; the slope gives phi = 33.1 degree, cohesion c = 0 (for sand).
- **Precaution** dry sand at same density in all tests; remove the pins before shearing; avoid vibration.
- Box area = 6 cm x 6 cm = 36 cm2.

Experiment 3e - Unconfined Compression Test (Cohesive Soil)

Aim

To determine the unconfined compressive strength qu and the undrained shear strength (cohesion) of cohesive soil. Reference: IS 2720 (Part 10).

Apparatus

- Compression machine (strain controlled) with proving ring, specimen of 38 mm dia and 76 mm height, sample extruder, split mould, vernier calliper, dial gauge, balance, stop watch.

Theory

In a UCS test the confining pressure is zero (sigma3 = 0). The specimen fails under the axial stress qu. For saturated clay phi = 0, so cohesion cu = qu / 2.

Formulas Axial strain e = delta L / L0. Corrected area A = A0 / (1 - e). Axial stress = P / A. Sensitivity St = qu (undisturbed) / qu (remoulded). Consistency :: qu below 25 kPa very soft, 25 to 50 soft, 50 to 100 medium, 100 to 200 stiff, above 200 very stiff.

Procedure

- Prepare the specimen by extruding the soil from the tube or by compacting in a mould; measure L0 and D0 and weigh it.
- Place it on the bottom plate, bring the upper plate in contact and set the proving ring and dial gauge to zero.
- Apply the load with a strain rate of 0.5 to 2 percent per minute and record the proving ring reading at every 0.5 or 1.0 mm deformation.
- Continue until failure (the load drops or the strain reaches 20 percent). Sketch the failure pattern.
- Take a sample from the failure zone for water content.

Observations

Deformation (mm)	Strain e	Proving ring reading (div)	Load P (N)	Corrected area A (mm2)	Stress (kPa)
1	0.0132	106	53.0	1149	46.1
2	0.0263	188	94.0	1165	80.7
3	0.0395	246	123.0	1181	104.2
4	0.0526	278	139.0	1197	116.1
5	0.0658	276	138.0	1214	113.7
6	0.0789	263	131.5	1231	106.8
7	0.0921	249	124.5	1249	99.7

Proving ring constant = 0.5 N per division; L0 = 76 mm, D0 = 38 mm, A0 = 1134 mm2.

Result

- **qu = 116.1 kPa, cu = qu / 2 = 58.1 kPa. Precaution** avoid disturbance of the sample; keep the strain rate constant; the ends of the sample should be flat.

Experiment 3f – Laboratory Vane Shear Test (Cohesive Soil)

Aim

To determine the undrained shear strength of soft cohesive soil by laboratory vane shear apparatus. Reference: IS 2720 (Part 30).

Apparatus

- Laboratory vane shear apparatus with a four bladed vane (12.5 mm dia x 12.5 mm height or 19 mm), torque spring with calibration chart, sample container (38 mm dia, 75 mm height), stop watch.

Theory

The vane is pushed into the soil and rotated; the torque needed to shear a cylindrical surface is measured.

Formula Shear strength s = T / [pi x (D squared H / 2 + D cubed / 6)] where T = torque at failure, D = diameter and H = height of the vane. Sensitivity = s (undisturbed) / s (remoulded).

Procedure

- Fill the sample in the cup, level it and place it on the base.
- Lower the vane into the soil to full depth (no disturbance).
- Rotate the handle at 0.1 degree per second (6 to 12 degree per minute) and record the torque until the soil fails (the reading falls).
- Note the angle of rotation at failure; convert to torque using the spring constant.
- Rotate the vane rapidly 10 times to remould and repeat to find the remoulded strength.

Observations and Result

Item	Value
Spring deflection at failure (degree)	41
Torque T (N mm)	114.8
Undrained strength s (kPa)	28.1
Remoulded strength (kPa)	10.3
Sensitivity	2.73

- **The test is used for soft clays only; Precaution** do not disturb the sample when inserting the vane.
- Data used: vane D = H = 12.5 mm, spring constant 2.8 N mm per degree, remoulded deflection 15 degree.

Experiment 3g – Tri-Axial Compression Test (Cohesionless Soil)

Aim

To determine the shear strength parameters of sand by tri-axial test (drained / consolidated drained). Reference: IS 2720 (Part 11 and Part 12).

Apparatus

- Tri-axial cell, loading frame, proving ring, pressure control system, rubber membrane, membrane stretcher, split mould (38 mm dia, 76 mm height), porous discs, burette for volume change, dial gauge, O-rings.

Theory

Specimen is enclosed in a rubber membrane and subjected to an all round cell pressure sigma3. The axial load is then raised until failure; the deviator stress is sigma1 - sigma3.

Formulas Deviator stress = P / A where A = A0 / (1 - e). Major principal stress sigma1 = sigma3 + deviator stress. For sand with c = 0, sin phi = (sigma1 - sigma3) / (sigma1 + sigma3). Mohr circles drawn for 3 tests; the common tangent gives phi.

Procedure

- Fix the membrane on the base pedestal with a stretcher and place the split mould; pour sand in layers at the required density and apply a small suction to hold the shape.
- Remove the mould, put the top cap, seal with O-rings and place the cell.
- Fill the cell with water, apply a cell pressure e.g. 1, 2 and 3 kg/cm2 in 3 different tests and allow drainage.
- Apply axial deformation at a slow rate (for drained test about 0.1 percent per minute) and record load and volume change.
- Continue to failure or 15 to 20 percent strain.

Observations

Test	Cell pressure sigma3 (kg/cm2)	Deviator stress at failure (kg/cm2)	Sigma1 = sigma3 + deviator
1	1.0	2.5	3.50
2	2.0	5.12	7.12
3	3.0	7.55	10.55

Result

- **Angle of shearing resistance phi = 33.9 degree (c = 0). Precaution** avoid membrane leakage and air bubbles; keep the rate of strain uniform.

Experiment 3h – California Bearing Ratio (CBR) Test

Aim

To determine the CBR value of soil for the design of flexible pavements.

Reference: IS 2720 (Part 16).

Apparatus

- CBR mould (150 mm dia, 175 mm height) with collar and perforated base, spacer disc, rammer, loading machine with plunger of 50 mm dia (1.25 mm per minute), proving ring, dial gauges, surcharge weights (2.5 kg each), swelling plate with tripod, water tank.

Theory

CBR is the ratio of the load needed to penetrate the soil by 2.5 mm or 5 mm to the load needed for the same penetration in standard crushed stone.

Formula CBR = (Test load / Standard load) x 100. Standard loads :: 1370 kg at 2.5 mm and 2055 kg at 5.0 mm penetration. The higher value is taken if the 5 mm CBR is more than the 2.5 mm value (repeat the test).

Procedure

- Prepare the soil at OMC and compact in the mould (3 layers with 56 blows of 2.6 kg rammer or 5 layers with 56 blows of 4.89 kg rammer, as required); remove the collar and trim.
- Soak the specimen in water for 96 hours (soaked CBR) with surcharge weights and measure the swell.
- Place the mould on the machine with surcharge weights and seat the plunger.
- Apply the load at 1.25 mm per minute and note the load at penetrations of 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0, 7.5, 10.0 and 12.5 mm.
- Plot load against penetration; apply the correction for the concave upward initial portion if needed.

Observations

Penetration (mm)	Dial reading (div)	Load (kg)
0.5	10	20
1.0	21	42
2.0	40	80
2.5	48	96
5.0	68	136

Result

- CBR at 2.5 mm = 7.0 %, at 5.0 mm = 6.6 %.
- **Use** CBR below 5 percent weak subgrade (needs treatment); 5 to 10 fair; above 10 good subgrade; above 80 good base course.
- **Precaution** compact uniformly; maintain the rate of penetration; keep the plunger seated.
- Proving ring constant = 2 kg per division. CBR at 2.5 mm (7.0 %) is greater than at 5 mm, so the design CBR = 7.0 %, a fair subgrade.

Summary of Important Formulas

Test	Key formula
Specific gravity	G = (W2 - W1) / [(W2 - W1) - (W3 - W4)]
Sieve analysis	Cu = D60/D10, Cc = D30 squared / (D60 x D10)
Plasticity index	Ip = wL - wP
Dry density	gamma_d = gamma / (1 + w)
Constant head k	k = Q L / (A h t)
Falling head k	k = 2.303 a L / (A t) x log (h1/h2)
Consolidation	cv = Tv Hdr squared / t
UCS	cu = qu / 2
Direct shear	tau = sigma tan phi
CBR	CBR = test load / standard load x 100

Viva Voce Questions

- What is the difference between bulk density and dry density?
- Why is the 25 blow water content taken as the liquid limit?
- What is Stokes law and where is it used?
- Define OMC and MDD. What is the zero air void line?
- Why is a falling head test used for clays?
- What is the effect of drainage in a shear test? Why is phi zero for saturated clay in UCS?
- What does CBR indicate and why is the sample soaked?
- Name two uses of consolidation test results.